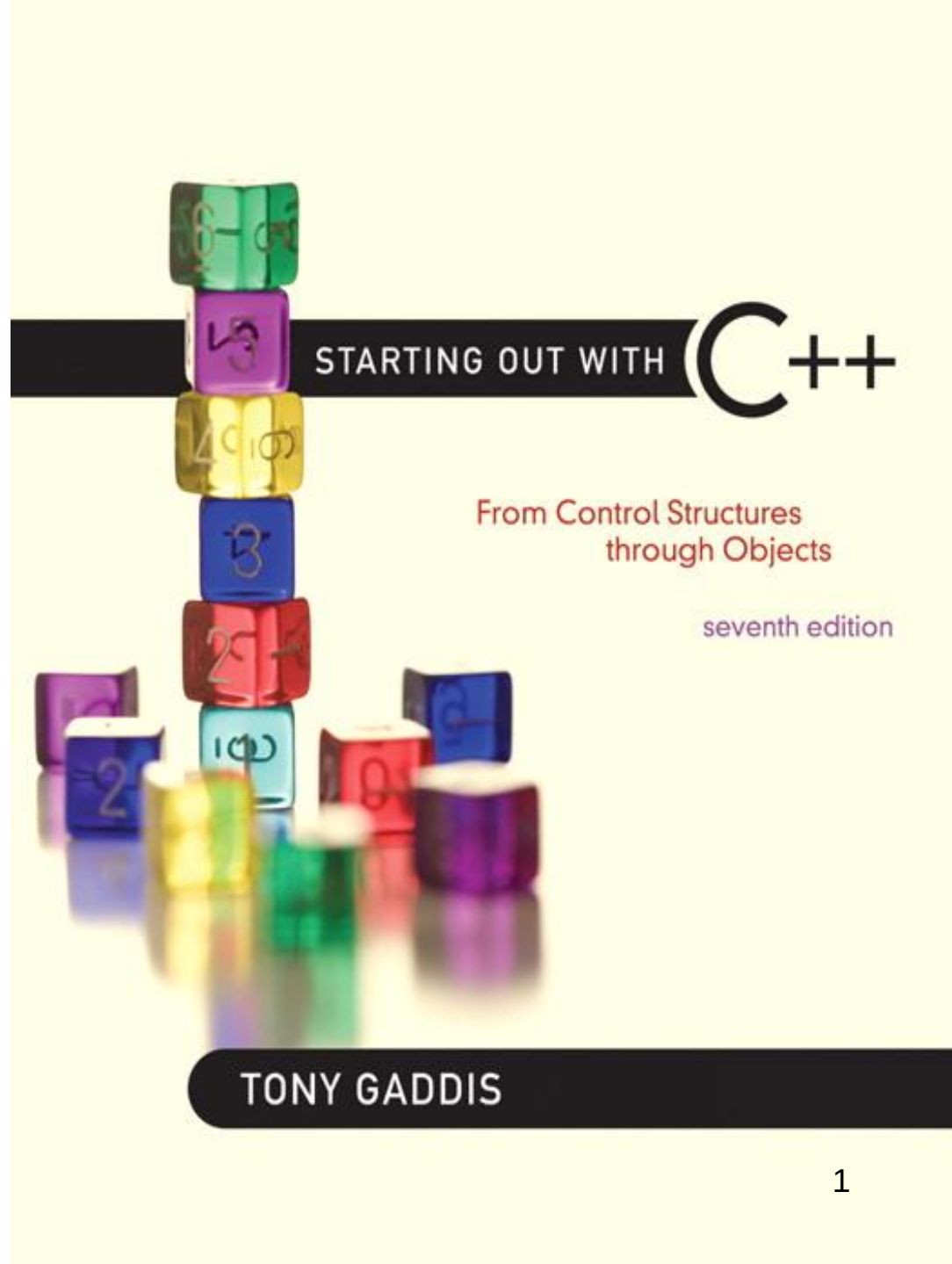


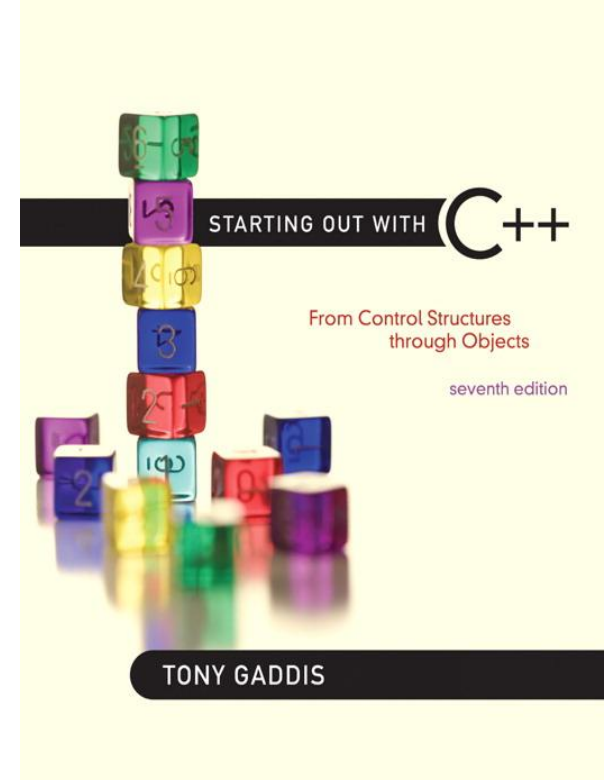
Chapter 1:

Introduction to Computers and Programming



1.1

Why Program?



Why Program?

Computer – programmable machine designed to follow instructions

Program – instructions in computer memory to make it do something

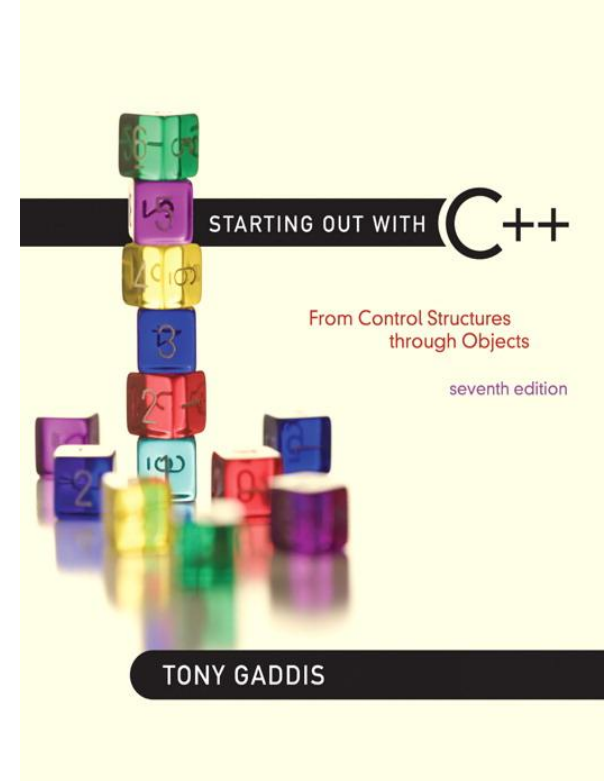
Programmer – person who writes instructions (programs) to make computer perform a task

SO, without programmers, no programs;
without programs, a computer cannot do anything

Programming is an art or a science?

- Both.
 - Art: program should be carefully designed.
 - Logical flow, mathematical procedures, user interface, user-friendliness,...
 - Science/Engineering: need many stages
 - testing, correction, redesign,...
 - Writing software demands discipline

1.2



Computer Systems: Hardware and Software

Main Hardware Component Categories:

1. Central Processing Unit (CPU)
2. Main Memory
3. Secondary Memory / Storage
4. Input Devices
5. Output Devices

Main Hardware Component Categories

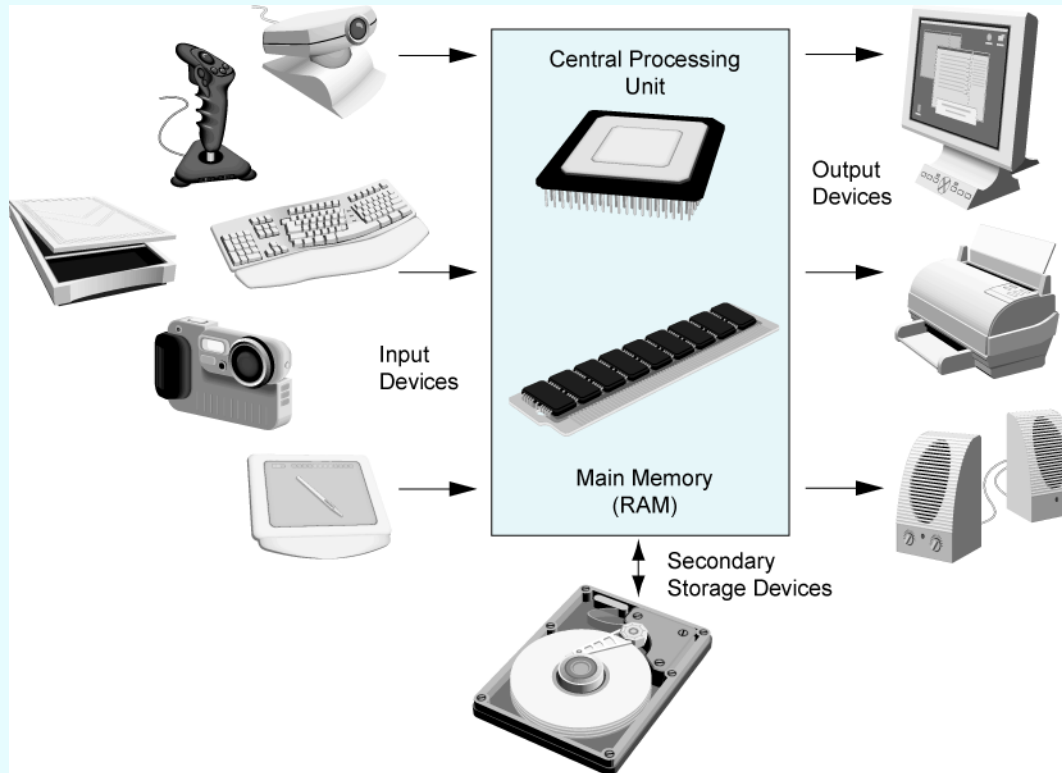
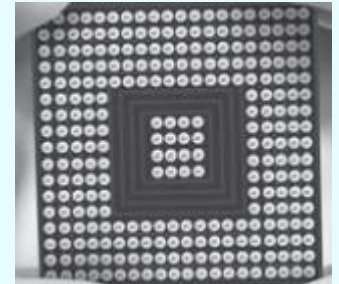
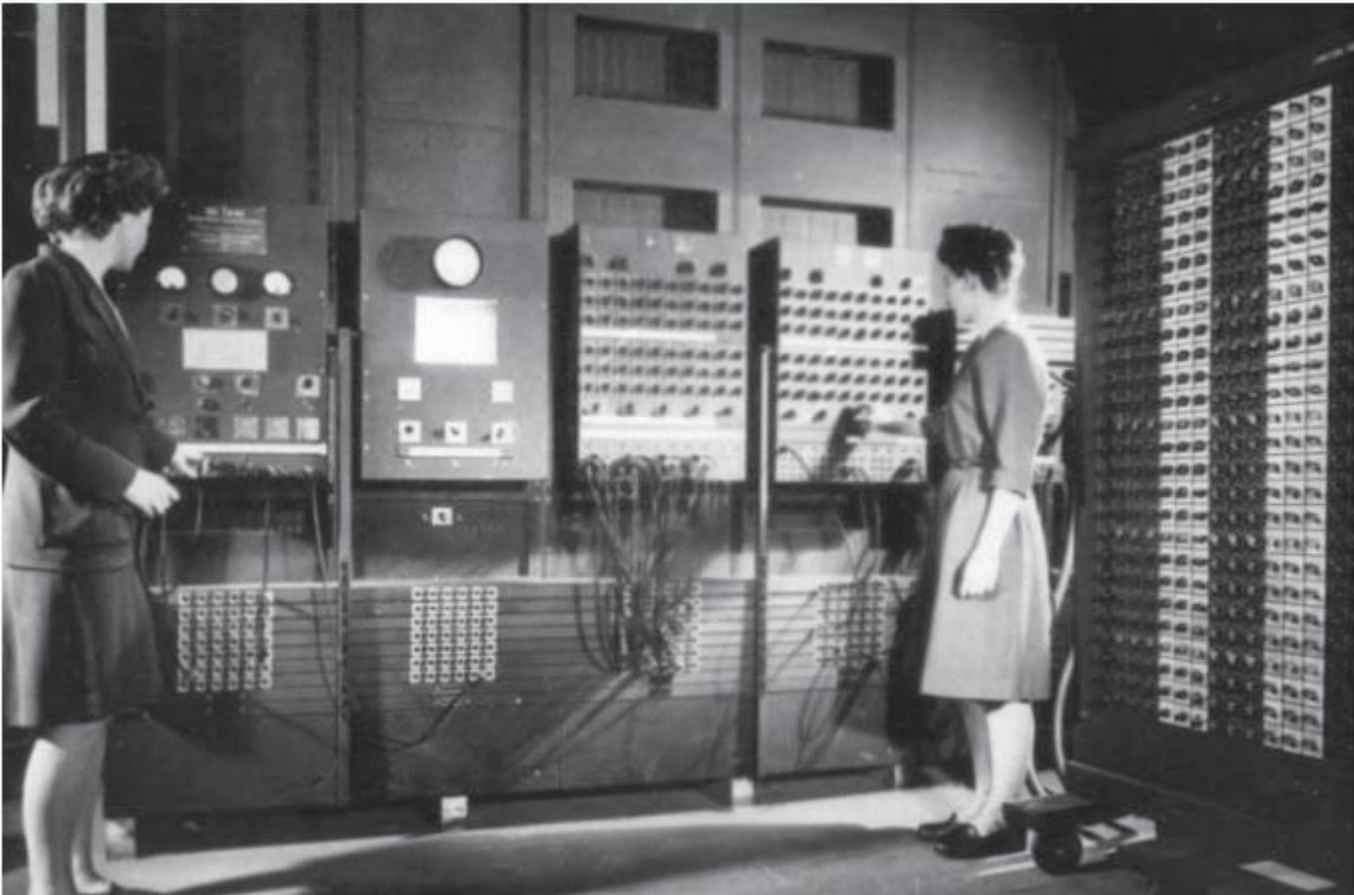


Figure 1-2

Central Processing Unit (CPU)



Central Processing Unit (CPU)

Comprised of:

Control Unit

- Retrieves and decodes program instructions

- Coordinates activities of all other parts of computer

Arithmetic & Logic Unit

- Hardware optimized for high-speed numeric calculation

- Hardware designed for true/false, yes/no decisions

CPU Organization

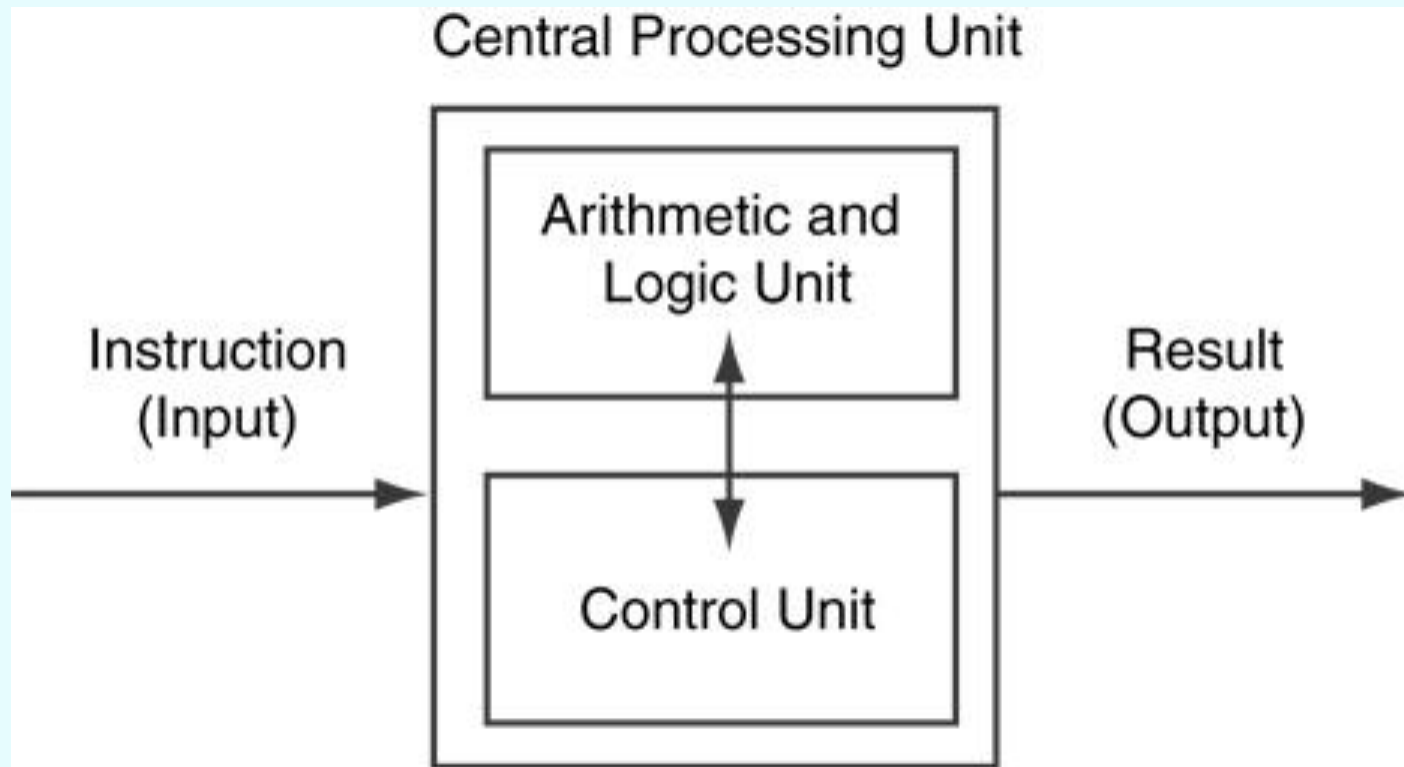


Figure 1-3

Main Memory

- It is volatile. Main memory is erased when program terminates or computer is turned off
- Also called Random Access Memory (RAM)
- Organized as follows:
 - bit: smallest piece of memory. Has values 0 (off, false) or 1 (on, true)
 - byte: 8 consecutive bits. Bytes have addresses.

Main Memory

- Addresses – Each byte in memory is identified by a unique number known as an *address*.

Main Memory

0	1	2	3	4	5	6	7	8	9
10	11	12	13	14	15	16	17	18	19
20	21	22	23	24	25	26	27	28	29

Diagram illustrating Main Memory layout (30 bytes, indexed 0 to 29). The memory is organized into three rows of 10 bytes each. The number 149 is stored at address 16, and the number 72 is stored at address 23.

- In Figure 1-4, the number 149 is stored in the byte with the address 16, and the number 72 is stored at address 23.

Secondary Storage

- Non-volatile: data retained when program is not running or computer is turned off
- Comes in a variety of media:
 - magnetic: floppy disk, hard drive
 - optical: CD-ROM, DVD
 - Flash drives, connected to the USB port

Input Devices

- Devices that send information to the computer from outside
- Many devices can provide input:
 - Keyboard, mouse, scanner, digital camera, microphone
 - Disk drives, CD drives, and DVD drives

Output Devices

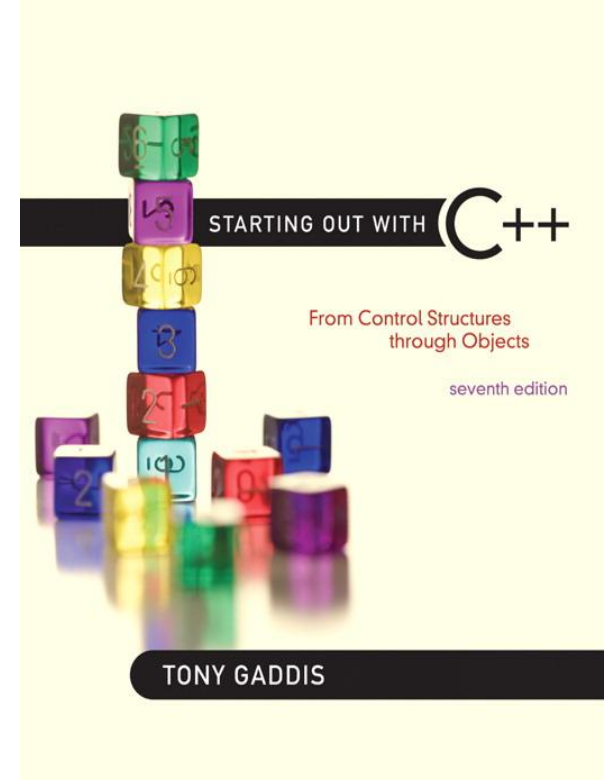
- Output – information the computer sends to the outside world.
- Output device format and present it.
- Some output devices:
 - monitor, printer, speaker, ..
 - Disk drives, CD drives, and DVD drives can also be considered output devices.

Software-Programs That Run on a Computer

- Categories of software:
 - System software: programs that manage the computer hardware and the programs that run on them. *Examples:* operating systems, utility programs, software development tools
 - Application software: programs that provide services to the user. *Examples :* word processing, games, programs to solve specific problems

1.3

Programs and Programming Languages



Programs and Programming Languages

- A program is a set of instructions that the computer follows to perform a task
- A programming language is a special language used to write computer programs.
- We start with an *algorithm*, which is a set of well-defined steps.

Example Algorithm for Calculating Gross Pay

1. Display a message on the screen asking “How many hours did you work?”
2. Wait for the user to enter the number of hours worked. Once the user enters a number, store it in memory.
3. Display a message on the screen asking “How much do you get paid per hour?”
4. Wait for the user to enter an hourly pay rate. Once the user enters a number, store it in memory.
5. Multiply the number of hours by the amount paid per hour, and store the result in memory.
6. Display a message on the screen that tells the amount of money earned. The message must include the result of the calculation performed in Step 5.

ALGORITHM IN REAL LIFE

- Consider the following ...

Problem: Baking a Cake

How to solve:

1. Start
2. Preheat the oven at 180°C
3. Prepare a baking pan
4. Beat butter with sugar
5. Mix them with flour, eggs and essence vanilla
6. Pour the dough into the baking pan
7. Put the pan into the oven
8. End

Machine Language

- Although the previous algorithm defines the steps for calculating the gross pay, it is not ready to be executed on the computer.
- The computer only executes *machine language* instructions

Machine Language

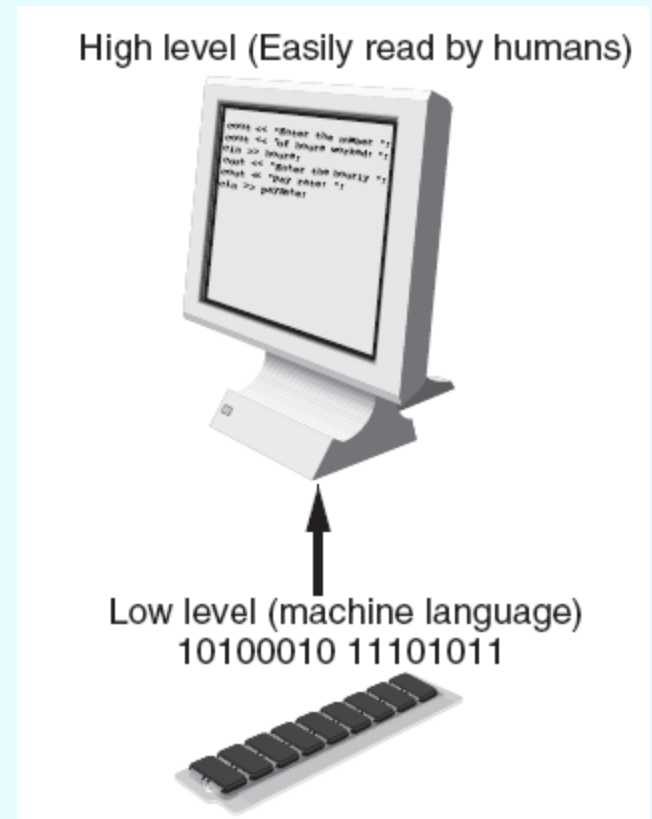
- Machine language instructions are binary numbers, such as

1011010000000101

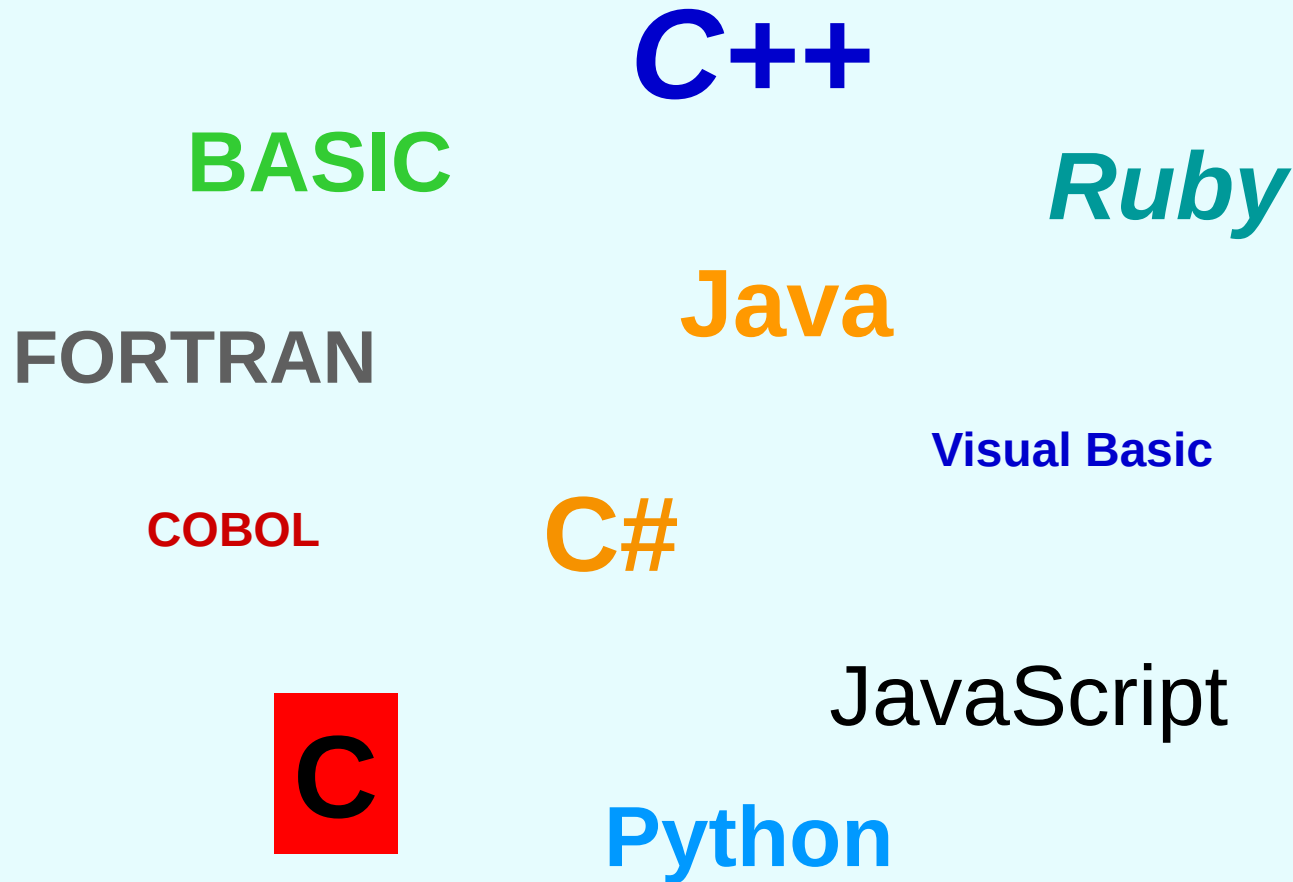
- Rather than writing programs in machine language, programmers use *programming languages*.

Programs and Programming Languages

- Types of languages:
 - Low-level: used for communication with computer hardware directly. Often written in binary machine code (0's/1's) directly.
 - High-level: closer to human language



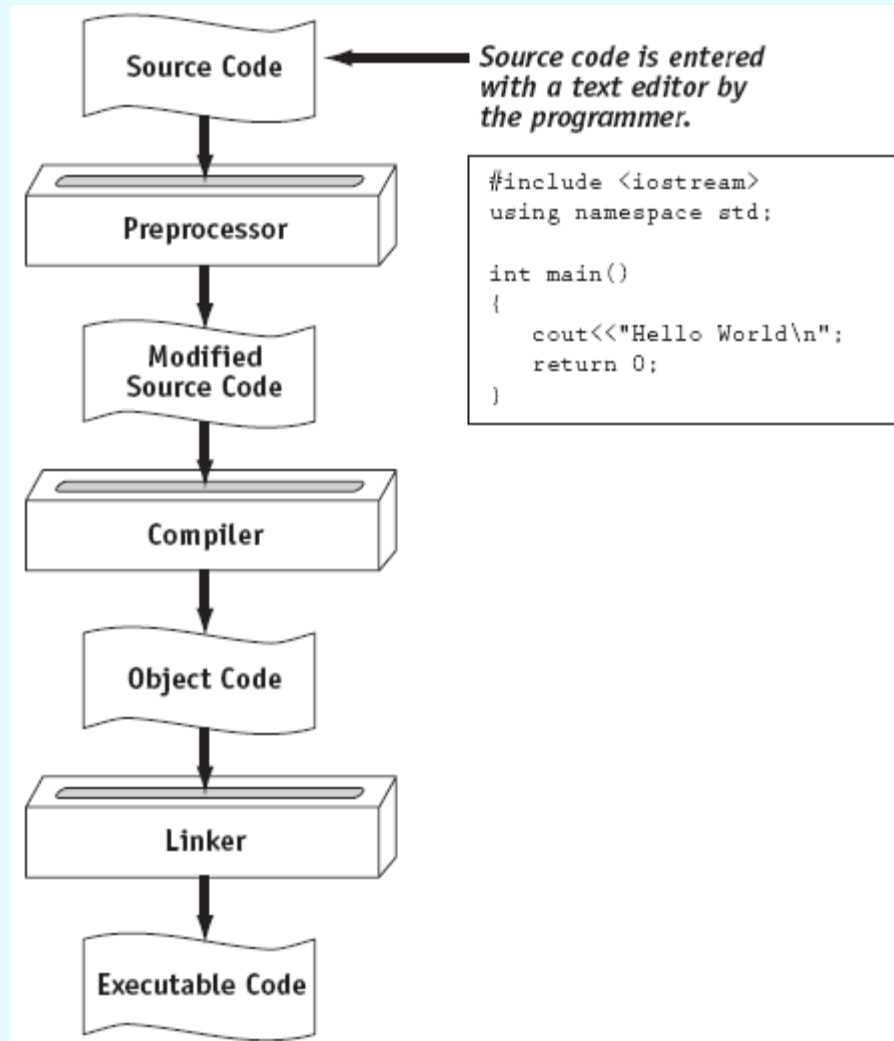
Some Well-Known Programming Languages (Table 1-1 on Page 10)



From a High-Level Program to an Executable File

- a) Create file containing the program with a text editor.
 - b) Run preprocessor to convert source file directives to source code program statements.
 - c) Run compiler to convert source program into machine instructions.
 - d) Run linker to connect hardware-specific code to machine instructions, producing an executable file.
- Steps b–d are often performed by a single command or button click.
 - Errors detected at any step will prevent execution of following steps.

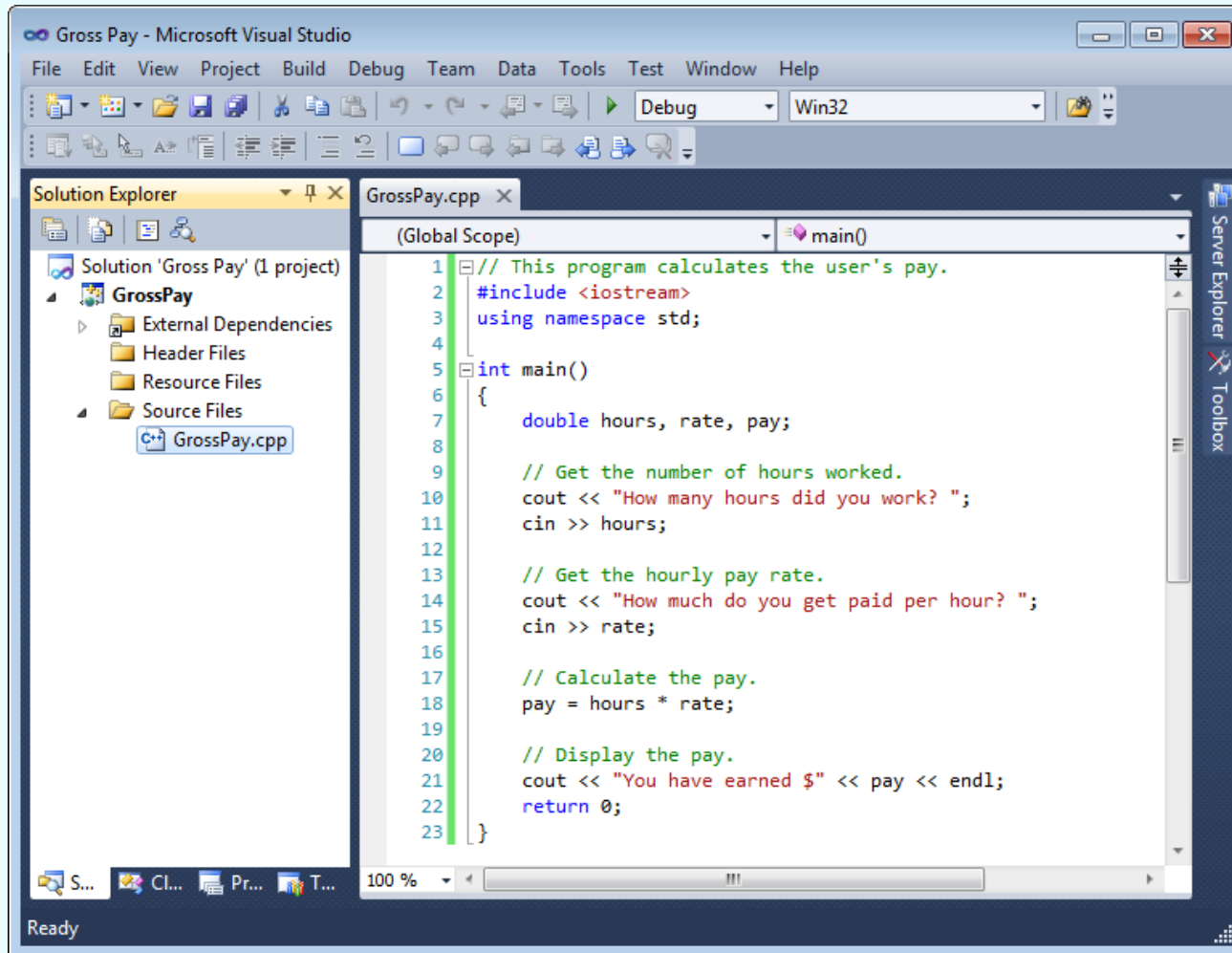
From a High-Level Program to an Executable File



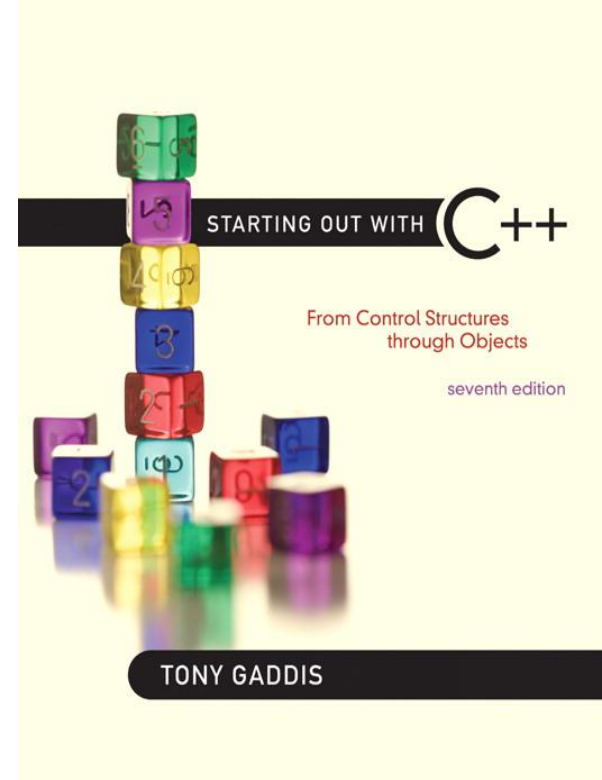
Integrated Development Environments (IDEs)

- An integrated development environment, or IDE, combine all the tools needed to write, compile, and debug a program into a single software application.
- Examples are Microsoft Visual C++, Turbo C++ Explorer, CodeWarrior, etc.

Integrated Development Environments (IDEs)



1.4



What is a Program Made of?

What is a Program Made of?

- Common elements in programming languages:
 - Key Words
 - Programmer-Defined Identifiers
 - Operators
 - Punctuation
 - Syntax

Program 1-1

```
1  // This program calculates the user's pay.
2  #include <iostream>
3  using namespace std;
4
5  int main()
6  {
7      double hours, rate, pay;
8
9      // Get the number of hours worked.
10     cout << "How many hours did you work? ";
11     cin >> hours;
12
13     // Get the hourly pay rate.
14     cout << "How much do you get paid per hour? ";
15     cin >> rate;
16
17     // Calculate the pay.
18     pay = hours * rate;
19
20     // Display the pay.
21     cout << "You have earned $" << pay << endl;
22     return 0;
23 }
```

Key Words

- Also known as reserved words
- Have a special meaning in C++
- Can not be used for any other purpose
- Key words in the Program 1-1: `using`, `namespace`, `int`, `double`, **and** `return`

Key Words

```
1  // This program calculates the user's pay.
2  #include <iostream>
3  using namespace std;
4
5  int main()
6  {
7      double hours, rate, pay;
8
9      // Get the number of hours worked.
10     cout << "How many hours did you work? ";
11     cin >> hours;
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13     // Get the hourly pay rate.
14     cout << "How much do you get paid per hour? ";
15     cin >> rate;
16
17     // Calculate the pay.
18     pay = hours * rate;
19
20     // Display the pay.
21     cout << "You have earned $" << pay << endl;
22     return 0;
23 }
```

Programmer-Defined Identifiers

- Names made up by the programmer
- Not part of the C++ language
- Used to represent various things: variables (memory locations), functions, etc.
- In Program 1-1: `hours`, `rate`, and `pay`.

Operators

- Used to perform operations on data
- Many types of operators:
 - Arithmetic - ex: $+$, $-$, $*$, $/$
 - Assignment – ex: $=$
- Some operators in Program1-1:
 $<< \gg = *$

Operators

```
1 // This program calculates the user's pay.
2 #include <iostream>
3 using namespace std;
4
5 int main()
6 {
7     double hours, rate, pay;
8
9     // Get the number of hours worked.
10    cout << "How many hours did you work? ";
11    cin >> hours;
12
13    // Get the hourly pay rate.
14    cout << "How much do you get paid per hour? ";
15    cin >> rate;
16
17    // Calculate the pay.
18    pay = hours * rate;
19
20    // Display the pay.
21    cout << "You have earned $" << pay << endl;
22    return 0;
23 }
```

Punctuation

- Characters that mark the end of a statement, or that separate items in a list
- In Program 1-1: , and ;

Punctuation

```
1 // This program calculates the user's pay.
2 #include <iostream>
3 using namespace std;
4
5 int main()
6 {
7     double hours, rate, pay;
8
9     // Get the number of hours worked.
10    cout << "How many hours did you work? ";
11    cin >> hours;
12
13    // Get the hourly pay rate.
14    cout << "How much do you get paid per hour? ";
15    cin >> rate;
16
17    // Calculate the pay.
18    pay = hours * rate;
19
20    // Display the pay.
21    cout << "You have earned $" << pay << endl;
22    return 0;
23 }
```

Syntax

- The rules of grammar that must be followed when writing a program
- Controls the use of key words, operators, programmer-defined symbols, and punctuation

Variables

- A variable is a named storage location in the computer's memory for holding a piece of data.
- In Program 1-1 we used three variables:
 - The **hours** variable was used to hold the hours worked
 - The **rate** variable was used to hold the pay rate
 - The **pay** variable was used to hold the gross pay

Variable Definitions

- To create a variable in a program you must write a variable definition (also called a variable declaration)
- Here is the statement from Program 1-1 that defines the variables:

```
double hours, rate, pay;
```

Variable Definitions

- There are many different types of data, which you will learn about in this course.
- A variable holds a specific type of data.
- The variable definition specifies the type of data a variable can hold, and the variable name.

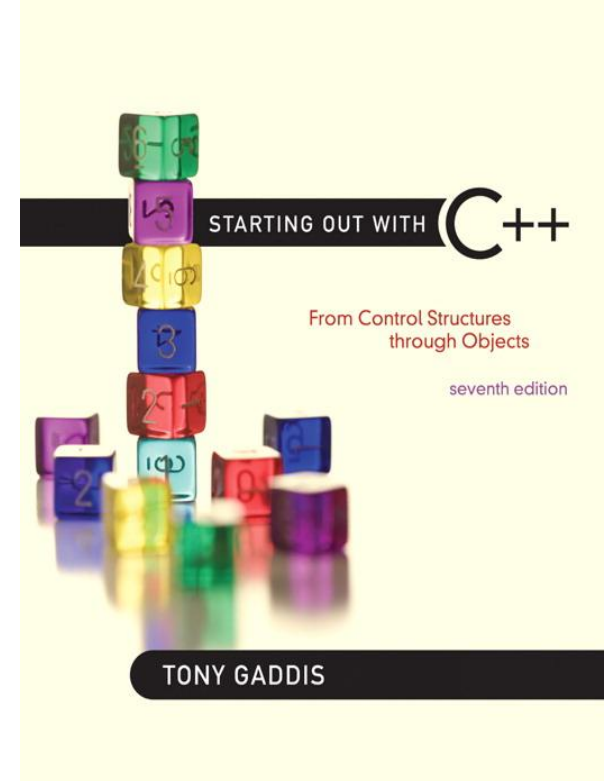
Variable Definitions

- Once again, line 7 from Program 1-1:

```
double hours, rate, pay;
```

- The word **double** specifies that the variables can hold double-precision floating point numbers. (You will learn more about that in Chapter 2)

1.5



Input, Processing, and Output

Input, Processing, and Output

Three steps that a program typically performs:

1) Gather input data:

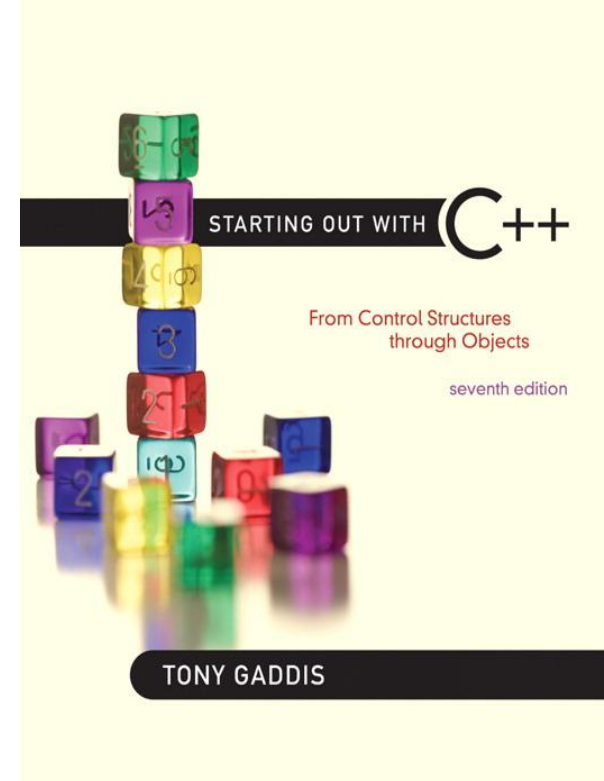
- from keyboard
- from files on disk drives

2) Process the input data

3) Display the results as output:

- send it to the screen
- write to a file

1.6



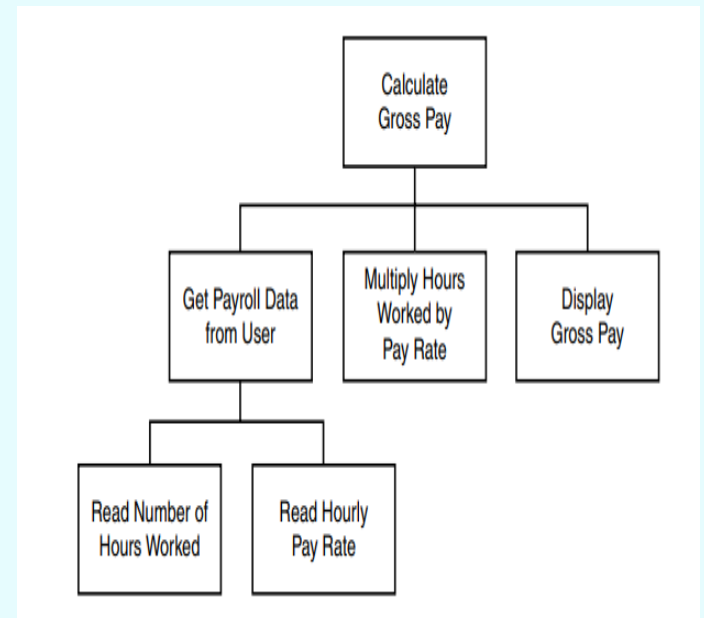
The Programming Process

The Programming Process

1. Clearly define what the program is to do.
2. Visualize the program running on the computer.
3. Use design tools such as a hierarchy chart, flowcharts, or pseudocode to create a model of the program.
4. Check the model for logical errors.
5. Type the code, save it, and compile it.
6. Correct any errors found during compilation. Repeat Steps 5 and 6 as many times as necessary.
7. Run the program with test data for input.
8. Correct any errors found while running the program. Repeat Steps 5 through 8 as many times as necessary.
9. Validate the results of the program.

HIERARCHY CHART

- Diagram that graphically depicts the structure of a program.
 - Boxes representing each step in the program.
Textually
 - connected in a way that illustrates their relationship to one another.



‘DIVIDE AND CONQUER’ STRATEGY IN ALGORITHM

Problem: Prepare a Breakfast

1. Start
- 2. Prepare a Breakfast**
3. End

‘DIVIDE AND CONQUER’ STRATEGY IN ALGORITHM

1. Start

2. Prepare a Breakfast

2.1 Prepare a tuna sandwich

2.2 Prepare some chips

2.3 Make a cup of coffee

3. End

‘DIVIDE AND CONQUER’ STRATEGY IN ALGORITHM

1. Start

2. Prepare a Breakfast

2.1 Prepare a tuna sandwich

2.1.1 Take 2 slices of bread

2.1.2 Prepare tuna paste

2.2 Prepare some chips

2.3 Make a cup of coffee

3. End

‘DIVIDE AND CONQUER’ STRATEGY IN ALGORITHM

1. Start

2. Prepare a Breakfast

2.1 Prepare a tuna sandwich

2.1.1 Take 2 slices of bread

2.1.2 Prepare tuna paste

2.2 Prepare some chips

2.2.1 Cut potatoes into slices

2.2.2 Fry the potatoes

2.3 Make a cup of coffee

3. End

‘DIVIDE AND CONQUER’ STRATEGY IN ALGORITHM

1. Start

2. Prepare a Breakfast

2.1. Prepare a tuna sandwich

2.1.1 Take 2 slices of bread

2.1.2 Prepare tuna paste

2.2. Prepare some chips

2.2.1 Cut potatoes into slices

2.2.2 Fry the potatoes

2.3. Make a cup of coffee

2.3.1 Boil water

2.3.2 Add water with sugar and coffee

3. End

HOW TO SPECIFY AN ALGORITHM?

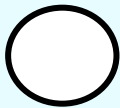
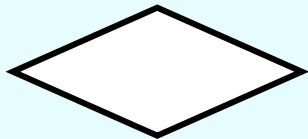
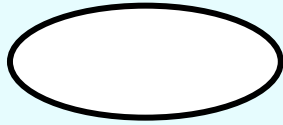
- An algorithm must be specific enough so that it can be conveniently translated into a computer program (using C++, for example).
- An algorithm can be specified:
 - Textually
For example, using pseudo code (see later)
 - Graphically
For example, using flowcharts or UML activity charts

FLOWCHARTS

- A flowchart is a graphical representation of the sequence of operations in a program.
- An algorithm can be represented graphically using a flowchart.

FLOWCHART NOTATIONS

Symbol



Semantic

Start/End

Process

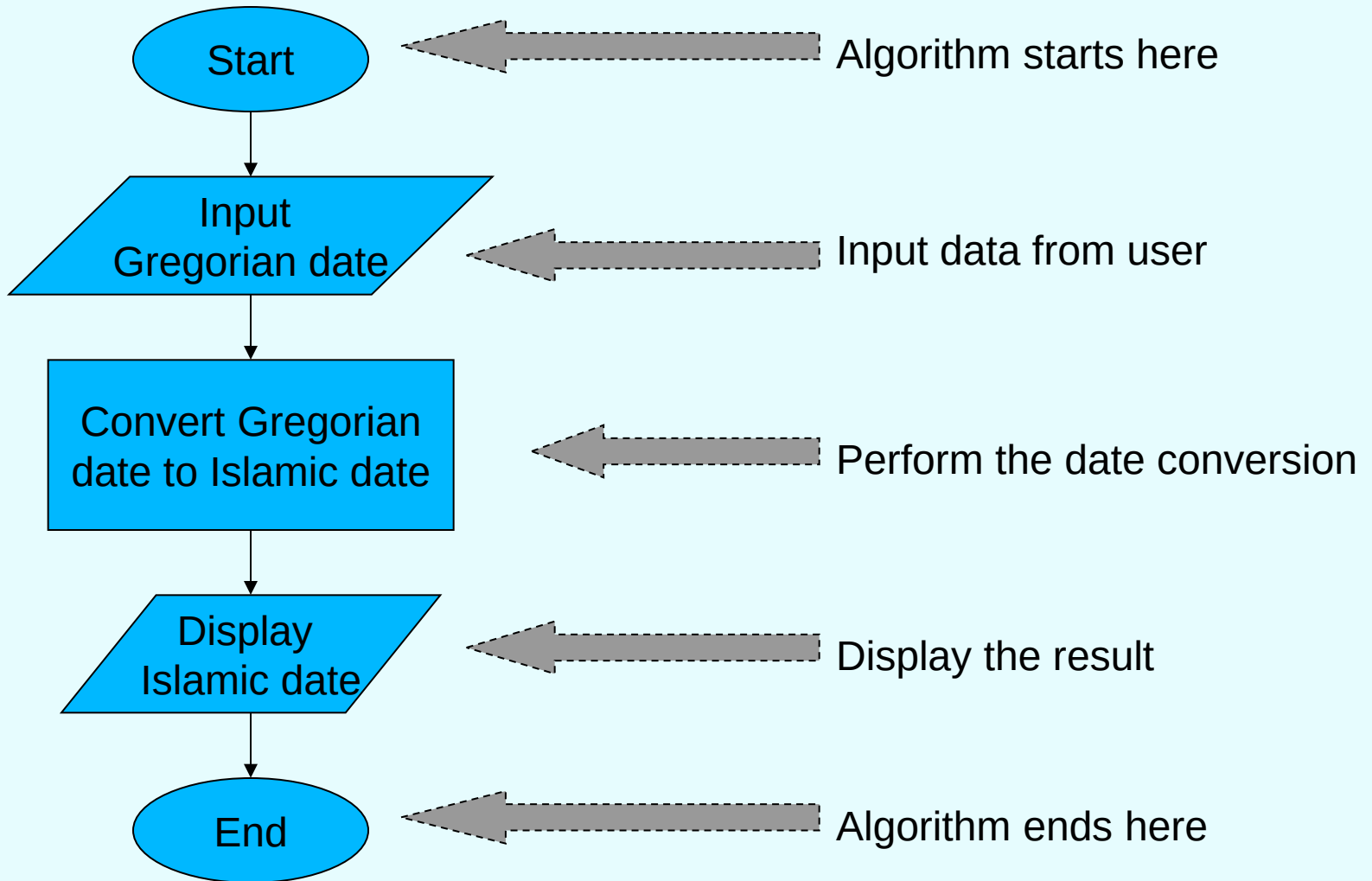
Input/Output

Test

Connector

Flow of activities

FLOWCHART: EXAMPLE 1



PSEUDOCODE

- An outline of a program, written in a form that can easily be converted into real programming statements. It resembles the actual program that will be implemented later. However, it cannot be compiled nor executed.
- Pseudocode normally codes the following actions:
 - Initialisation of variables
 - Assignment of values to the variables
 - Arithmetic operations
 - Relational operations

EXAMPLE OF PSEUDOCODE

1. Start
2. Read `quantity`
3. Read `price_per_kg`
4. `price` $\leftarrow$ `quantity` * `price_per_kg`
5. Print `price`
6. End

SYNTAX ERRORS

- Syntax errors are errors in the source code which are related to the syntax of the language.
- Syntax errors are detected by the compiler. An executable file will be generated by the compiler only if the source code it compiles has no syntax errors.
- Syntax errors are reported by the compiler in the form of error messages.

#include <iostream>

using namespace std;

int main() {

cout << "This program has errors

return;

} Error messages
displayed by the
compiler

```
sufian@fujitsu3 /cygdrive/d/PalmOScygwin/sufian
$ g++ prog.cpp
prog.cpp:4:17: missing terminating " character
prog.cpp: In function `int main()':
prog.cpp:5: error: syntax error before `return'
prog.cpp:6:2: warning: no newline at end of file

sufian@fujitsu3 /cygdrive/d/PalmOScygwin/sufian
$
```

LOGICAL ERRORS

- Logical errors are errors which are related to program logic.
- Normally, logical errors are not detectable by the compiler.
- Logical errors are usually detected during program runtime. For example, a program producing unexpected results is an indication that it has logical errors.
- It is important to remember that if the compiler does not produce any error messages, it does not mean that your program is free of logical errors.

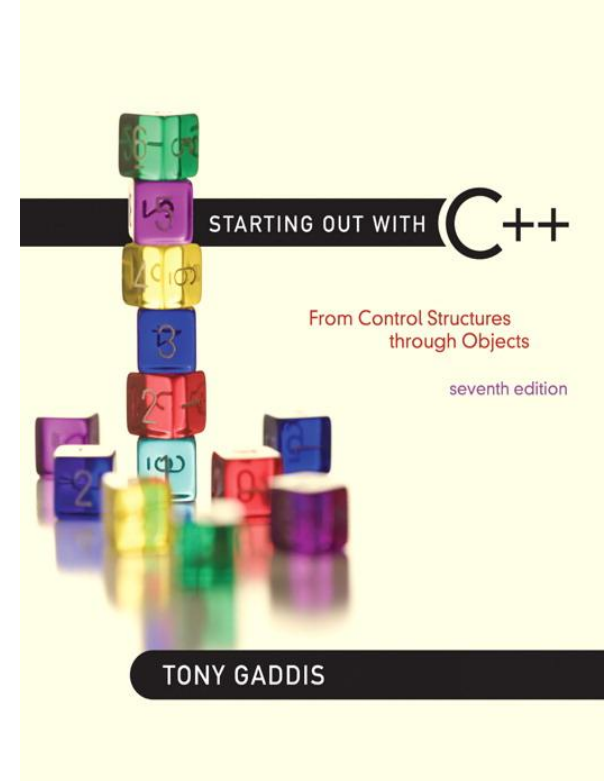
LOGICAL ERRORS

- Possible to remove all syntax errors in a program and still not have it run
- Even if it runs, it may still not do what you meant it to do
- For example,

$2 + 3 * 5$ and $(2 + 3) * 5$

are both syntactically correct expressions,
but have different meanings

1.7



Procedural and Object-Oriented Programming

Procedural and Object-Oriented Programming

- Procedural programming: focus is on the process. Procedures/functions are written to process data.
- Object-Oriented programming: focus is on objects, which contain data and the means to manipulate the data. Messages sent to objects to perform operations.

First week homework

- Starting out C++ Tony Gaddis (6eds)
 - Check point: 1.7, 1.10, 1.12, 1.15, 1.17
 - 30 (Algorithm workbench), 34 (Predict the result)
- Starting out C++ Tony Gaddis (7eds)
 - Programming challenges: Candy Bar Sales.